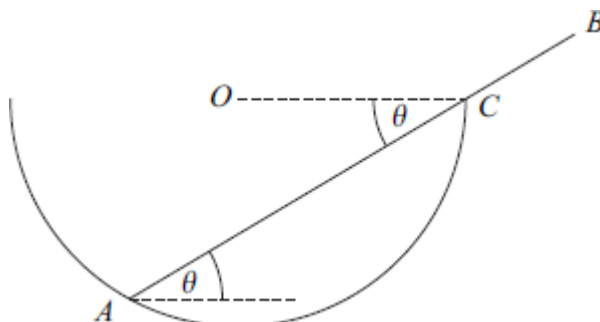


Q1.

A smooth hollow hemisphere, of radius a and centre O , is fixed so that its rim is in a horizontal plane. A smooth uniform rod AB , of mass m , is in equilibrium, with one end A resting on the inside of the hemisphere and the point C on the rod being in contact with the rim of the hemisphere. The rod, of length l , is inclined at an angle θ to the horizontal, as shown in the diagram.



- (a) Explain why the reaction between the rod and the hemisphere at point A acts through O .

(1)

- (b) Draw a diagram to show the forces acting on the rod.

(2)

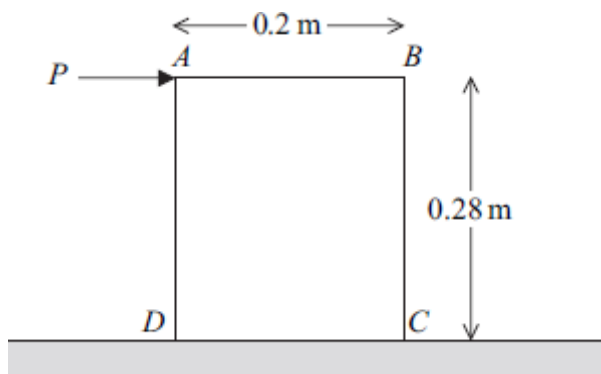
- (c) Show that $l = \frac{4a \cos 2\theta}{\cos \theta}$.

(5)

(Total 8 marks)

Q2.

A box of breakfast cereal, of total mass 0.4 kg , rests on a rough horizontal plane. It can be assumed that the centre of the mass of the box with its contents is on a vertical line through the centre of the box. The diagram shows a vertical cross section $ABCD$ through the centre of mass of the box and its contents, where $AB = 0.2 \text{ m}$ and $BC = 0.28 \text{ m}$. A horizontal force of magnitude P is applied at A , in the direction of AB .



The coefficient of friction between the box and the plane is μ . As P is gradually increased from zero, the box slides before it topples if, and only if, $\mu < k$. Find the value of k .

Q3.

Four forces act at the points with coordinates as listed below.

Force		Point
$2\mathbf{i} + 3\mathbf{j}$	acts at	$A(1, 0, 0)$
$2\mathbf{i} - 3\mathbf{j}$	acts at	$B(1, 1, 0)$
$-\mathbf{j} + 2\mathbf{k}$	acts at	$C(0, 0, 1)$
$\mathbf{j} + 2\mathbf{k}$	acts at	$D(0, 1, 1)$

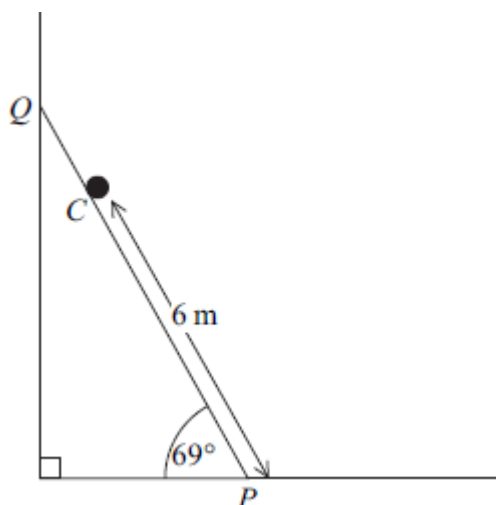
- (a) Find the resultant of the four forces. (1)
- (b) Find the total moment of the four forces about O . (5)
- (c) An additional force $\mathbf{P}$ acts at a point on the y -axis so that the five forces are in equilibrium.
- Determine the coordinates of the point on the y -axis at which $\mathbf{P}$ acts. (6)

(Total 12 marks)

Q4.

A uniform ladder PQ , of length 8 metres and mass 28 kg, rests in equilibrium with its foot, P , on a rough horizontal floor and its top, Q , leaning against a smooth vertical wall. The vertical plane containing the ladder is perpendicular to the wall and the angle between the ladder and the floor is 69° .

A man, of mass 72 kg, is standing at the point C on the ladder so that the distance PC is 6 metres. The man may be modelled as a particle at C .



- (a) Draw a diagram to show the forces acting on the ladder. (2)
- (b) With the man standing at the point C , the ladder is on the point of slipping.
- (i) Show that the magnitude of the reaction between the ladder and the vertical wall is 256 N, correct to three significant figures. (4)
- (ii) Find the coefficient of friction between the ladder and the horizontal floor. (4)
- (Total 10 marks)

Q5.

A light rod has its ends at the points $P(-2, -1, 4)$ and $Q(4, 1, 6)$. A force $\mathbf{F}$ acts at the point M , the mid-point of PQ , where $\mathbf{F} = a\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

- (a) Show that $\overrightarrow{PM} = 3\mathbf{i} + \mathbf{j} + \mathbf{k}$. (2)
- (b) Find the moment of $\mathbf{F}$ about the point P , giving your answer in terms of a . (3)
- (c) Given that the magnitude of the moment is $5\sqrt{2}$, find the two possible values of a . (4)
- (Total 9 marks)

Q6.

Four forces, acting in the x - y plane, act at the points with coordinates as listed below, where p and q are constants and $\mathbf{i}$ and $\mathbf{j}$ are parallel to the x -axis and y -axis respectively.

Force	Coordinate
$-2\mathbf{i}$	acting at $(3, 3)$
$-5\mathbf{j}$	acting at $(4, 0)$
$p\mathbf{i} + \mathbf{j}$	acting at $(-8, -2)$
$-3\mathbf{i} - 4\mathbf{j}$	acting at $(0, q)$

- (a) Write down the resultant of these forces in terms of p . (1)
- (b) (i) Given that the system of forces is equivalent to a single force, $\mathbf{F}$, acting parallel to the y -axis, find the value of p . (2)
- (ii) Given further that the equation of the line of action of $\mathbf{F}$ is $x = 3$, find the value

of q .

(6)

- (c) The system of forces can be replaced by a single force acting at the origin together with a couple C .

Write down the magnitude of C and indicate its sense on a diagram.

(2)

(Total 11 marks)

Q7.

Three forces, $-\mathbf{i} + \mathbf{j}$, $4\mathbf{i} - 2\mathbf{k}$ and $-3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, act at the points whose coordinates are $(0, 2, 1)$, $(3, -1, 0)$ and $(4, 0, -5)$ respectively.

- (a) Find the resultant of these three forces.

(1)

- (b) Find the moment of these three forces about $(0, 0, 0)$.

(6)

- (c) Explain why these three forces are equivalent to a couple.

(2)

(Total 9 marks)

Q8.

The forces $\begin{bmatrix} a \\ -3 \end{bmatrix}$, $\begin{bmatrix} -4 \\ 2a \end{bmatrix}$ and $\begin{bmatrix} -8 \\ 0 \end{bmatrix}$ act in the x - y plane at the points with coordinates $(2, 6)$, $(3, -5)$ and $(-1, 2)$ respectively.

- (a) In the case where this system of forces is equivalent to a single force acting at the origin O and a couple C , find the magnitude of C and deduce that it is independent of a .

(5)

- (b) In the case where $a = 4$, this system of forces is equivalent to a single force $\mathbf{F}$ acting at the point $(d, 0)$.

- (i) Find $\mathbf{F}$.

(2)

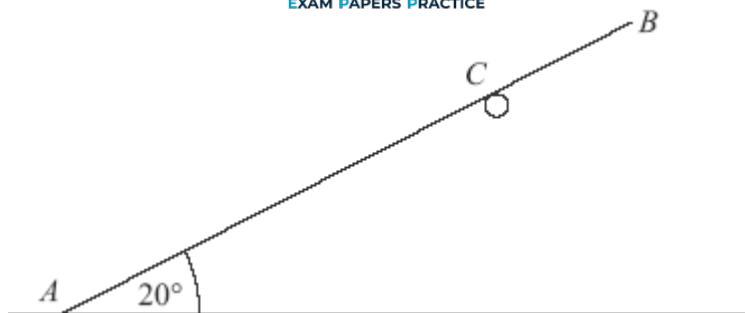
- (ii) Find d .

(2)

(Total 9 marks)

Q9.

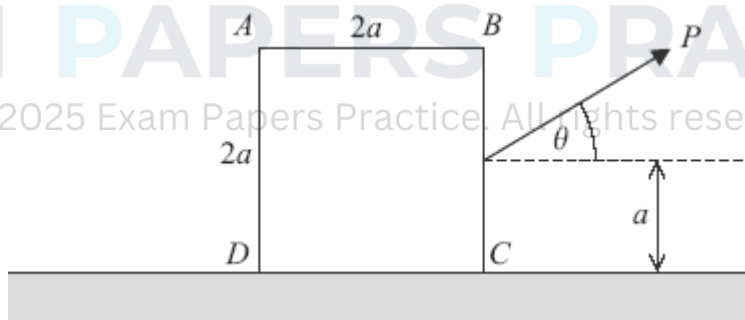
A uniform rod AB , of length 4 m and mass 6 kg, rests in equilibrium with one end, A , on smooth horizontal ground. The rod rests on a rough horizontal peg at the point C , where AC is 3 m. The rod is inclined at an angle of 20° to the horizontal.



- (a) Draw a diagram to show the forces acting on the rod. (2)
- (b) Find the magnitude of the normal reaction force between the rod and the ground. (3)
- (c) (i) Find the normal reaction acting on the rod at C. (5)
 (ii) Find the friction force acting on the rod at C.
- (d) In this position, the rod is on the point of slipping. Calculate the coefficient of friction between the rod and the peg. (2)
- (Total 12 marks)**

Q10.

A uniform cube, of side $2a$ and mass m , rests on a rough horizontal plane. The diagram shows a vertical cross-section $ABCD$ through the centre of mass of the cube.



A force, of magnitude P , is applied at the mid-point of BC . This force acts in the plane $ABCD$ and makes an angle θ with the horizontal. The coefficient of friction between the cube and the plane is μ .

- (a) In the case where the cube does not slide but is on the point of toppling about the edge through C , find an expression for P in terms of m , g and θ . (3)
- (b) In the case where the cube remains upright but is on the point of sliding along the plane, show that $P = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$.

(4)

- (c) Find an inequality that μ must satisfy if the cube slides before it topples.

(3)

- (d) Would your answer in part (c) change if the mass of the cube were doubled? Explain why.

(2)

(Total 12 marks)

Q11.

Two forces, $2\mathbf{i} + a\mathbf{k}$ and $-2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$, act at the points whose coordinates are $(1, 0, 3)$ and $(-1, 2, 0)$ respectively.

- (a) Show that the resultant moment of these forces about the origin is $6\mathbf{i} + (9 - a)\mathbf{j} + 3\mathbf{k}$.

(5)

- (b) This system is equivalent to a force $\mathbf{F}$ that acts at the origin together with a couple of magnitude 7.

- (i) Show that one possible value of a is 7 and find the other possible value of a .

(4)

- (ii) In the case where $a = 7$, find $\mathbf{F}$.

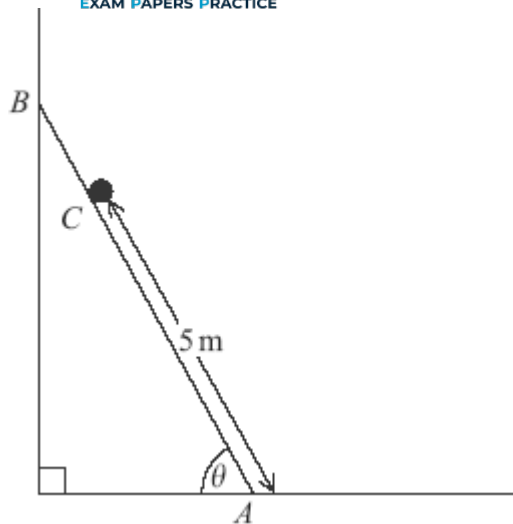
(2)

(Total 11 marks)

Q12.

A uniform ladder, of length 6 metres and mass 22 kg, rests with its foot, A , on a rough horizontal floor and its top, B , leaning against a smooth vertical wall. The vertical plane containing the ladder is perpendicular to the wall, and the angle between the ladder and the floor is θ .

A man, of mass 90 kg, is standing at point C on the ladder so that the distance AC is 5 metres. With the man in this position, the ladder is on the point of slipping. The coefficient of friction between the ladder and the horizontal floor is 0.6. The man may be modelled as a particle at C .



- (a) Show that the magnitude of the frictional force between the ladder and the horizontal floor is 659 N, correct to three significant figures.

(4)

- (b) Find the angle θ .

(5)

(Total 9 marks)

Q13.

The forces $\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$, $2\mathbf{i} - 7\mathbf{j} - \mathbf{k}$ and $4\mathbf{j} - 2\mathbf{k}$ act at the points with coordinates (2, 1, 0), (1, 13, -2) and (6, 4, -7) respectively. The resultant of the three forces is a single force $\mathbf{F}$.

- (a) Show that the magnitude of $\mathbf{F}$ is 7.

(3)

- (b) The point P has coordinates (3, 6, -3).

- (i) Find the moment of the force $4\mathbf{j} - 2\mathbf{k}$ about P .

(4)

- (ii) Given that the resultant of the two forces $\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ and $2\mathbf{i} - 7\mathbf{j} - \mathbf{k}$ acts through P , state the moment of $\mathbf{F}$ about P , giving a reason for your answer.

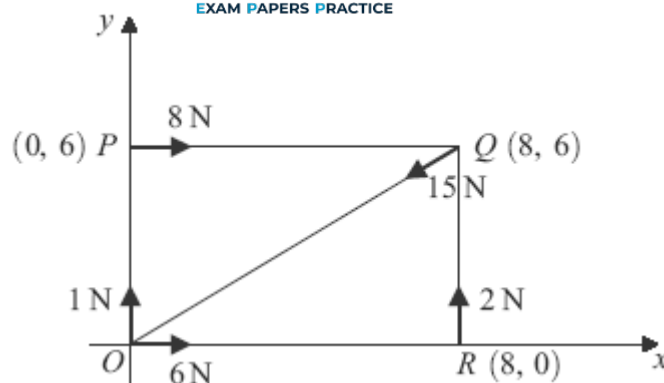
(2)

(Total 9 marks)

Q14.

The points O , P , Q and R have coordinates (0, 0), (0, 6), (8, 6) and (8, 0) respectively. The units of length are metres.

- A force of 1 N acts at O along OP .
- A force of 2 N acts at R along RQ .
- A force of 6 N acts at O along OR .
- A force of 8 N acts at P along PQ .
- A force of 15 N acts at Q along QO .

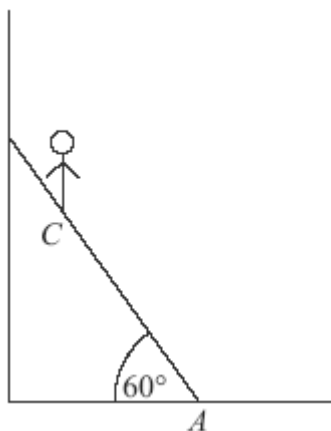


- (a) Show that the resultant of the five forces has magnitude $2\sqrt{10}$ N. (5)
- (b) An anticlockwise couple of magnitude 20 N m together with these five forces is equivalent to a single force of magnitude $2\sqrt{10}$ which has a line of action passing through the point $(0, d)$.
- (i) Find d . (5)
- (ii) Determine the equation of the line of action of the resultant, giving your answer in vector form. (3)
- (Total 13 marks)

Q15.

A uniform ladder of length 4 metres and mass 20 kg rests in equilibrium with its foot, A, on a rough horizontal floor and its top leaning against a smooth vertical wall. The vertical plane containing the ladder is perpendicular to the wall and the angle between the ladder and the floor is 60° .

A man of mass 80 kg is standing at point C on the ladder. With the man in this position, the ladder is on the point of slipping. The coefficient of friction between the ladder and the floor is 0.4. The man may be modelled as a particle at C.



- (a) Draw a diagram to show the forces acting on the ladder. (2)

- (b) Show that the magnitude of the frictional force between the ladder and the ground is 392 N.

(3)

- (c) Find the distance AC .

(6)

(Total 11 marks)

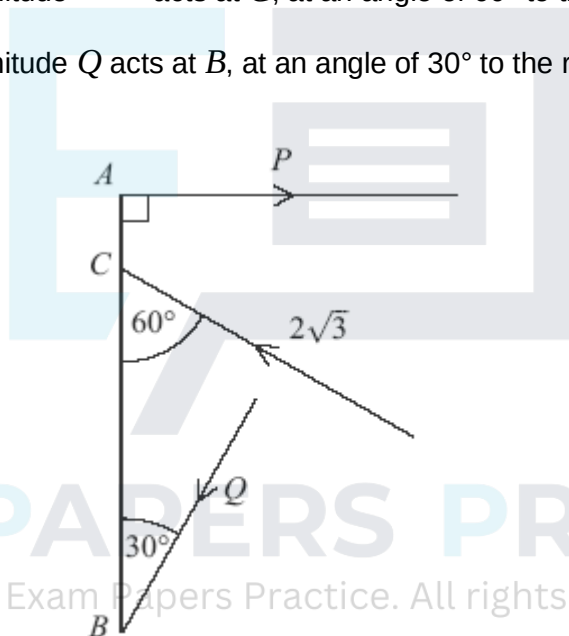
Q16.

A light rod AB has length 5 metres and the point C on the rod is 1 metre from A . The rod is on a smooth horizontal table and is acted upon by three horizontal forces of magnitude P , Q and $2\sqrt{3}$ newtons.

The force of magnitude P acts at A , at right angles to the rod.

The force of magnitude $2\sqrt{3}$ acts at C , at an angle of 60° to the rod.

The force of magnitude Q acts at B , at an angle of 30° to the rod, as shown in the diagram.



The three forces are equivalent to a couple.

- (a) Show that $Q = 2$ and find the value of P .

(5)

- (b) Determine the magnitude of the couple.

(3)

- (c) State the sense of the couple.

(1)

(Total 9 marks)

Q17.

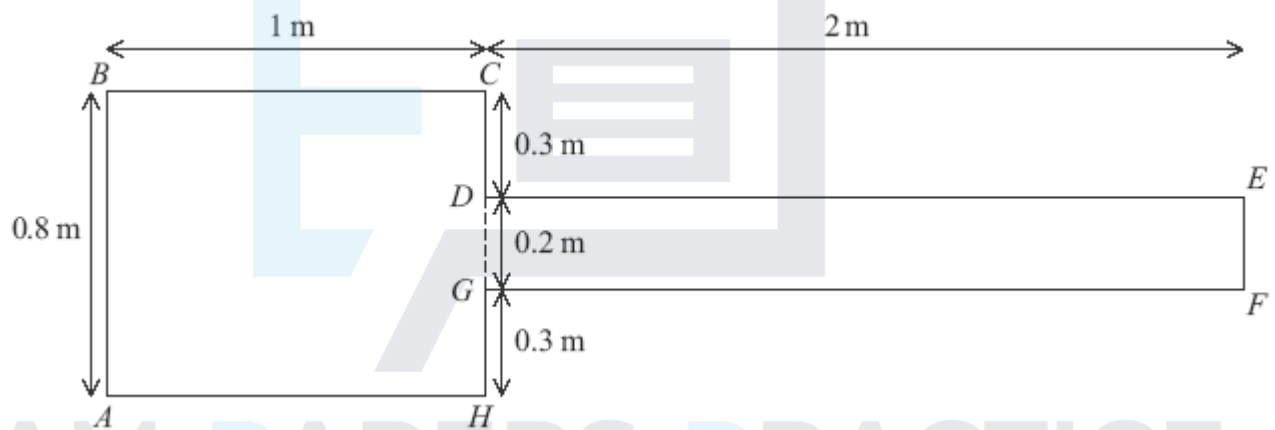
A light rod has its ends at the points $A(2, 3, 5)$ and $B(4, 6, -1)$. A force $\mathbf{F}$ acts at B , where

$$\mathbf{F} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$$

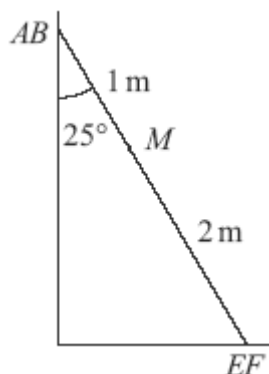
- (a) Find $\overrightarrow{AB}$. (1)
- (b) Find the moment of $\mathbf{F}$ about the point A . (3)
- (c) Show that the magnitude of this moment is $10\sqrt{5}$. (2)
- (d) Hence, or otherwise, find the acute angle between $\mathbf{F}$ and the rod, giving your answer to the nearest degree. (4)
- (Total 10 marks)

Q18.

A sign advertising a gym consists of two rectangles $ABCH$ and $DEFG$ fixed rigidly together. The sign can be modelled as a uniform lamina, as shown in the diagram.



- (a) The centre of mass of the sign is at the point M . Show that M lies on the line CH . (4)
- (b) The sign is placed with its side EF on rough horizontal ground and its side AB against a smooth vertical wall. The sign rests in equilibrium at an angle of 25° with the **vertical**, as shown in the diagram.



The weight of the sign is 90 newtons.

- (i) By taking moments, show that the normal reaction force between the sign and the wall is $60 \tan 25^\circ$ newtons. (3)
- (ii) The coefficient of friction between the sign and the ground is μ .

Show that $\mu \geq \frac{2}{3} \tan 25^\circ$.

(4)

(Total 11 marks)

Q19.

Two forces, $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$, act at the points $P(1, -1, 6)$ and $Q(0, 3, -2)$ respectively. These two forces together with a force $\mathbf{F}$, acting through the point Q , form a couple.

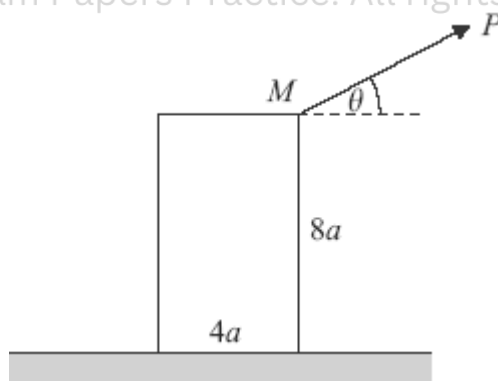
- (a) (i) Find the force $\mathbf{F}$. (3)
- (ii) Show that the magnitude of $\mathbf{F}$ is $3\sqrt{10}$. (2)
- (b) Find the moment of the couple. (5)

(Total 10 marks)

Q20.

A uniform block in the shape of a cuboid has weight W , a square base of side $4a$, a height of $8a$ and stands on a rough horizontal surface. The coefficient of friction between the block and the surface is μ . A rope is attached to the point M , the midpoint of a top edge of the block.

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The rope is pulled with a force P , which acts at an angle of θ above the horizontal and is perpendicular to the top edge.

- (a) Find P , in terms of W and θ , if the block is on the point of toppling. (4)

(4)

- (b) Show that $P = \frac{\mu W}{\cos \theta + \mu \sin \theta}$ if the block is on the point of sliding. (7)
- (c) Given that $\tan \theta = 1$, find an inequality that μ must satisfy if the block slides before it topples. (5)
- (Total 16 marks)

Q21.

Three forces $2\mathbf{i}$, $3\mathbf{i} - 5\mathbf{j} + a\mathbf{k}$ and $b\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ act at the points with coordinates $(1,1,0)$, $(0, 0, 0)$ and $(-1, 2, 1)$ respectively, where a and b are constants.

Given that the three forces form a couple, find:

- (a) the values of a and b ; (2)
- (b) the moment of the couple. (4)
- (Total 6 marks)

Q22.

Forces $\begin{bmatrix} 8 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 6 \\ 5 \end{bmatrix}$, $\begin{bmatrix} -2 \\ -2 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$ act at the points with coordinates $(0,0)$, $(0,3)$, $(3,4)$

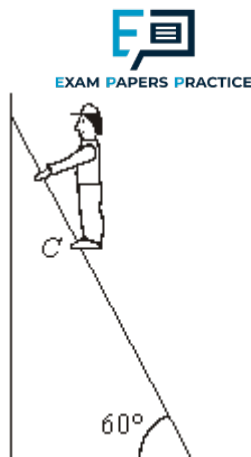
and $(4,0)$ respectively.

- (a) (i) Find the magnitude of the resultant $\mathbf{F}$ of this system of forces. (3)
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- (ii) Show that the line of action of $\mathbf{F}$ cuts the y -axis at the point $(0,2)$. (4)
- (b) The system of forces is equivalent to a force acting at the origin together with a couple C . Write down the magnitude of C and indicate its sense on a diagram. (2)
- (Total 9 marks)

Q23.

A uniform ladder of length 3 metres and mass 35 kg rests in equilibrium with its foot on a rough horizontal floor and its top leaning against a smooth vertical wall. The vertical plane containing the ladder is perpendicular to the wall and the angle between the ladder and the floor is 60° .

A man of mass 90 kg is standing at a point C on the ladder. With the man in this position, the ladder is on the point of slipping. The coefficient of friction between the ladder and the floor is 0.3. The man may be modelled as a particle at C .

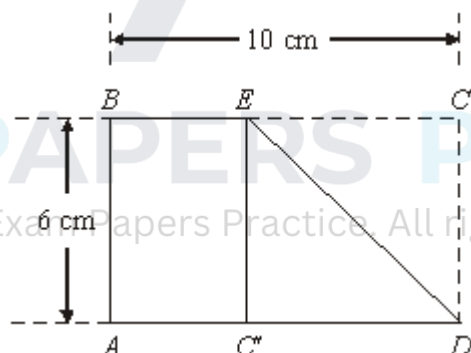


- (a) Draw a diagram to show the forces acting on the ladder. (1)
- (b) Find the magnitude of the frictional force between the ladder and the ground. (3)
- (c) Show that the man is just over half way up the ladder. (6)
- (Total 10 marks)

Q24.

A uniform rectangular lamina $ABCD$ has sides $AB = 6$ cm and $BC = 10$ cm.

The section ECD is folded along the line DE so that DC lies along DA and the point C is at C' , as shown in the diagram.



- (a) (i) Show that the centre of mass of the triangular section $EC'D$ of the new body is 6 cm from AB . (2)
- (ii) Hence show that the centre of mass of the new body is 4.4 cm from AB . (2)
- (iii) Find the distance of the centre of mass of the new body from AD . (2)
- (b) The body is now freely suspended from B and is then rotated in the vertical plane of the body to the position where AD is horizontal, as in the diagram above. It is held in equilibrium in this position by a couple of magnitude K .

Given that the weight of the body is 2 N, find the value of K .

(2)
(Total 8 marks)

Q25.

Forces $4\mathbf{i} + 5\mathbf{j}$, $2\mathbf{i} - \mathbf{j}$ and $-3\mathbf{i} + 2\mathbf{j}$ act through the points $(1, 2)$, $(3, -1)$ and $(-2, 1)$ respectively.

(a) The three forces are equivalent to the single force $\mathbf{P}$. Show that $\mathbf{P} = 3\mathbf{i} + 6\mathbf{j}$.

(1)

(b) (i) Show that the forces have a total moment of magnitude 5 about the origin.

(4)

(ii) State the sense of this total moment.

(1)

(c) The line of action of $\mathbf{P}$ cuts the y -axis at the point $(0, d)$. Find the value of d .

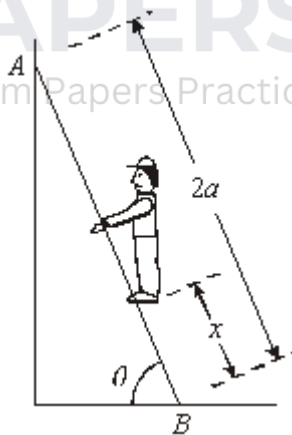
(3)

(Total 9 marks)

Q26.

A uniform ladder AB , of weight W_1 and length $2a$, rests on rough horizontal ground, with A against a smooth vertical wall. The ladder makes an angle θ with the horizontal where $\tan \theta = 3$.

A man, of weight W_2 , is standing at a distance x from the bottom of the ladder. The system is in equilibrium. Assume that the man's centre of mass is directly above his feet and that any contact forces through his hands are negligible.



(a) Draw a diagram showing all the forces acting on the ladder.

(2)

(b) Find, in terms of W_1 , W_2 , a and x , the force of the wall on the ladder at A .

(4)

(c) Given that the coefficient of friction between the ladder and the ground is $\frac{3}{10}$,

show that

$$x \leq \frac{\alpha(4W_1 + 9W_2)}{5W_2}$$

(4)

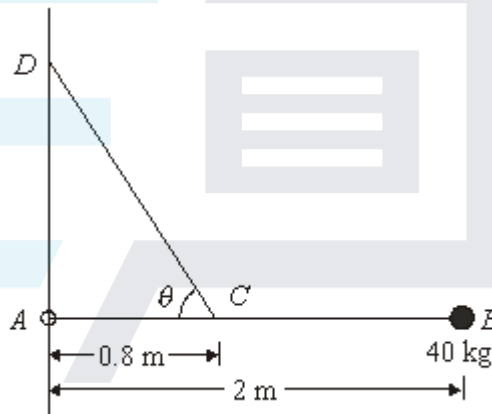
- (d) Deduce that the man can stand at the top of the ladder without it slipping, provided that $W_2 \leq 4W_1$.

(3)

(Total 13 marks)

Q27.

A uniform horizontal beam AB has mass 50 kg and length 2 metres. The end A of the beam is attached by a smooth hinge to a vertical wall. The beam is maintained in a horizontal position by a light inextensible rope, which is attached to the beam at the point C , 0.8 metres from A , and to the wall at the point D , directly above A . The rope is at an angle θ to the horizontal. A mass of 40 kg is attached to the beam at B .



- (a) Draw a diagram to show the forces acting on the beam.

(2)

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- (b) The breaking strength of the rope is 2000 N. Show that angle θ must be at least 53° , correct to the nearest degree.

(6)

(Total 8 marks)

Q28.

Forces $\begin{bmatrix} 2 \\ -8 \end{bmatrix}$, $\begin{bmatrix} a \\ b \end{bmatrix}$ and $\begin{bmatrix} -3 \\ -4 \end{bmatrix}$ act at points $(0, 0)$, $(3, 2)$ and $(4, -2)$ respectively.

The three forces are equivalent to a single force $\begin{bmatrix} 3 \\ -6 \end{bmatrix}$.

- (a) Find the values of a and b .

(3)

- (b) (i) Find the magnitude of the resultant moment of the forces about the origin.

(4)

(ii) State the sense of this moment.

(1)

(c) The line of action of the single equivalent force cuts the y -axis at point $(0, d)$. Find the value of d .

(2)

(Total 10 marks)

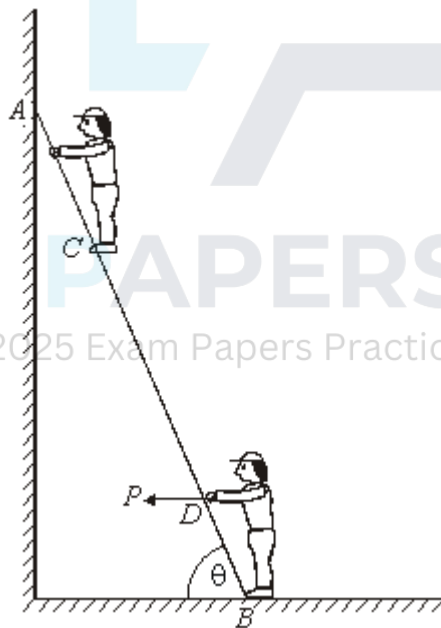
Q29.

A uniform ladder AB , of weight W and length $4a$, is resting with A against a smooth vertical wall and B on rough horizontal ground.

A man of weight $4W$ is standing at a point C on the ladder, where $BC = 3a$. The force exerted by the man's hands on the ladder is negligible and the centre of mass of the man is vertically above C .

Another man stands on the ground near the bottom of the ladder and pushes on the ladder at the point D , where $BD = a$. The force applied at D is horizontal, of magnitude P and is just sufficient to prevent the ladder sliding away from the wall.

The ladder is inclined at an angle θ $\tan^{-1}\left(\frac{12}{5}\right)$ to the horizontal.



The coefficient of friction between the base of the ladder and the ground is $\frac{1}{4}$.

(a) Show that the horizontal frictional force at B is of magnitude $\frac{5W}{4}$.

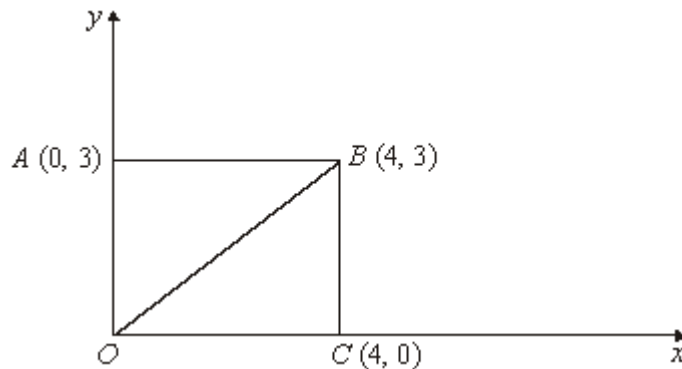
(3)

(b) Find the value of P in terms of W .

(6)

Q30.

The points O , A , B and C have coordinates $(0, 0)$, $(0, 3)$, $(4, 3)$ and $(4, 0)$ respectively.



A clockwise couple of magnitude 19 N m acts in the plane together with forces of magnitudes 5 N , 6 N , 4 N , 7 N and 5 N acting along OA , BA , CB , OC , and OB respectively.

- (a) Show that the resultant of this system of forces and the couple is of magnitude 13 N .

(5)

- (b) (i) Show that the line of action of the resultant cuts the y -axis at $(0, -3)$.

(5)

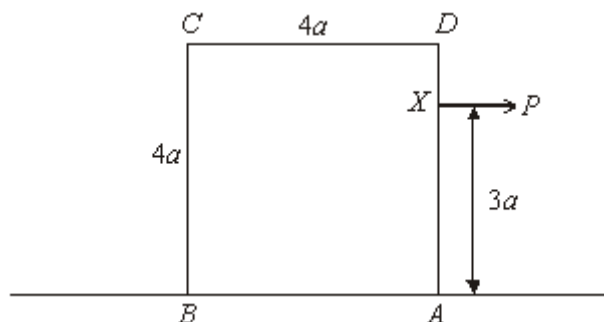
- (ii) Find the equation of the line of action of the resultant.

(3)

(Total 13 marks)

Q31.

A uniform cube, of side $4a$ and weight W , rests on a horizontal rough plane surface. The diagram shows a vertical cross-section $ABCD$ parallel to opposite faces of the cube and through its centre of mass.



A horizontal force P is applied to AD at a point X where $AX = 3a$. The coefficient of friction between the cube and the plane is μ .

- (a) In the case where the cube remains upright but slides along the plane under the action of P , determine an inequality for P in terms of W and μ .

(3)

- (b) In the case where the cube does not slide under the action of P but is on the point of toppling about the edge through A :

(i) draw a diagram showing the lines of action of the forces on the cube;

(1)

(ii) determine the value of P in terms of W .

(2)

- (c) The value of P is gradually increased from zero. If $\mu = 0.6$, determine whether the cube will slide before it topples.

(3)

(Total 9 marks)

Q32.

The diagram shows a uniform rod AB of length $2a$ and weight W . The end A is freely hinged to a fixed support. A horizontal force $2W$ is applied at B to keep the rod in equilibrium as shown in the diagram below.



- (a) Write down the horizontal and vertical components of the reaction, P , of the hinge on the rod at A .

(2)

- (b) Find the value of the angle θ which the rod makes with the downward vertical.

(3)

- (c) With the aid of a diagram, describe the position of one point, other than A , through which the line of action of P must pass.

(2)

(Total 7 marks)

Q33.

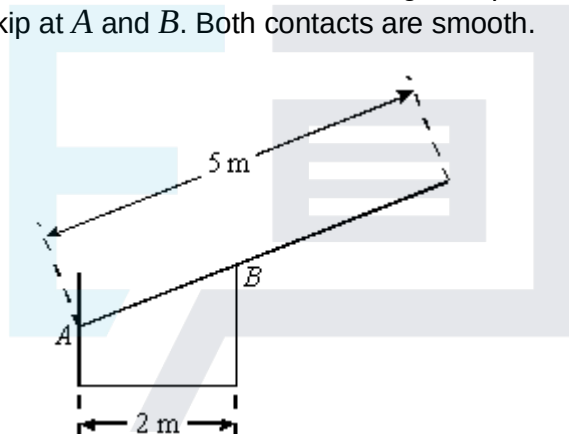
Forces $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 3 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ act in the x - y plane at the points with coordinates $(0, 0)$, $(0, 2)$, $(3, 4)$ and $(3, 0)$ respectively.

- (a) Find the magnitude of the resultant $\mathbf{F}$ of this system of forces. (4)
- (b) The line of action of $\mathbf{F}$ cuts the x -axis at the point $(d, 0)$. By taking moments about the origin, or otherwise, find d . (4)
- (c) The system of forces is equivalent to a force $\mathbf{F}$ acting at the origin, together with a couple L . Write down the magnitude of L and indicate its sense on a diagram. (2)
- (Total 10 marks)

Q34.

As part of an initiative test, a youth group is attempting to balance a uniform rigid pole, of weight 500 N and of length 5 m, in a vertical sided rubbish skip of width 2 m.

The diagram shows a vertical cross section through the pole and skip. The pole is in contact with the skip at A and B . Both contacts are smooth.

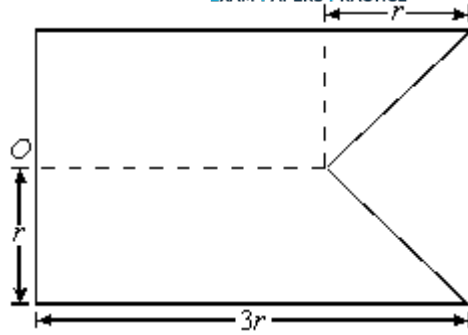


- (a) Show on a diagram the forces acting on the pole. (1)
- (b) Show that, when the pole is in equilibrium, it makes an angle of approximately 22° with the horizontal. (8)

(8)
(Total 9 marks)

Q35.

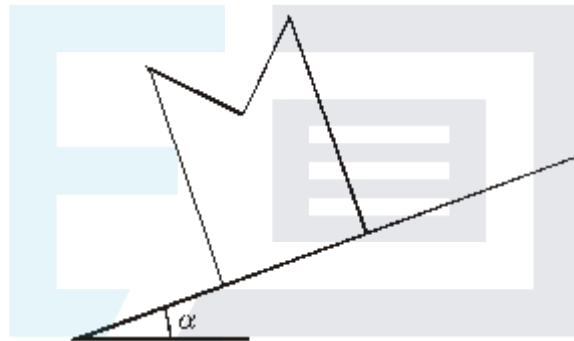
A solid, uniform cylinder, of radius r and length $3r$, has a cone of radius r and height r drilled out of one end. The diagram shows a cross section along the axis of symmetry of the body. Point O is the centre of the plane circular surface of the body.



- (a) Show that the centre of mass of the body is $\frac{43r}{32}$ from point O .

(6)

- (b) The body is placed with its plane surface on an adjustable rough slope as shown. The angle α which the plane makes with the horizontal is gradually increased from zero.



Assuming that the body does **not** slide, find, to the nearest degree, the value of angle α when the body just begins to topple.

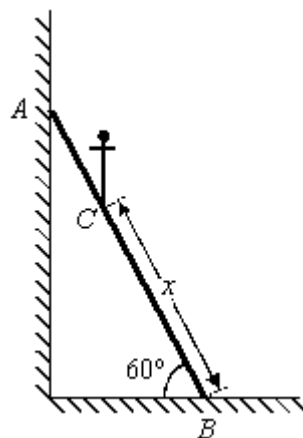
(3)

(Total 9 marks)

Q36.

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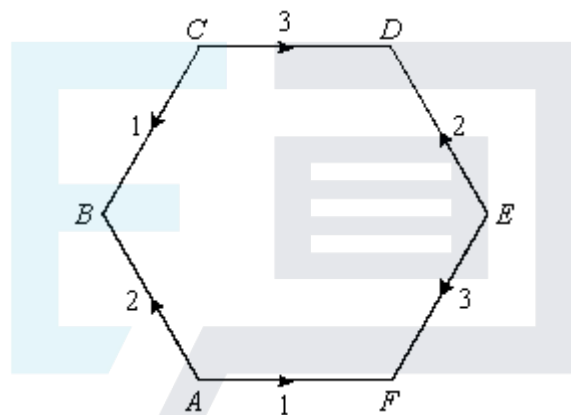
A uniform ladder AB , of mass 20 kg and length 5 metres, rests with B on a rough horizontal floor and with A against a smooth vertical wall. The ladder is at an angle of 60° to the horizontal and the coefficient of friction between the ladder and the floor is 0.4. A man of mass 80 kg is standing on the ladder at the point C , x metres from B .



- (a) Draw a diagram showing the forces acting on the ladder. (2)
- (b) Given that the ladder is in limiting equilibrium:
- (i) show that the force in the ladder at A is of magnitude 392 N; (4)
- (ii) find the value of x . (5)
- (Total 11 marks)

Q37.

The diagram shows a regular hexagon $ABCDEF$ with forces 2 N along AB , 1 N along CB , 3 N along CD , 2 N along ED , 3 N along EF and 1 N along AF .



- (a) Show that the resultant of these six forces is zero. (5)
- (b) The six forces form a couple, L .
- (i) Given that the length of each side of the hexagon is a metres, find, in terms of a , the magnitude of the couple L . (3)
- (ii) State the sense of the couple L . (1)

(Total 9 marks)

Q38.

A uniform rod AB has length 4 m. Three particles are attached to the rod as listed below.

Mass of particle	Distance from A
3 kg	1 m
12 kg	2.5 m

15 kg	3 m
-------	-----

The mass of the rod is 10 kg.

- (a) Show that the centre of mass of the rod and the particles is 2.45 metres from A .

(4)

- (b) Ropes are attached to each end of the rod. Find the tension in each rope if the rod is supported in equilibrium by the ropes, with the rod horizontal and the ropes vertical.

(5)

(Total 9 marks)

Q39.

Two forces, $8\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $-4\mathbf{i} + 2\mathbf{j} + 14\mathbf{k}$, act at the points whose coordinates are $(2, -1, 4)$ and $(7, -3, 2)$ respectively. The three unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are mutually perpendicular.

The resultant of these forces, together with a force $\mathbf{F}$ which acts through the origin, form a couple.

Find:

- (a) (i) the force $\mathbf{F}$;

(2)

- (ii) the magnitude of $\mathbf{F}$;

(2)

- (b) the moment of the couple.

(5)

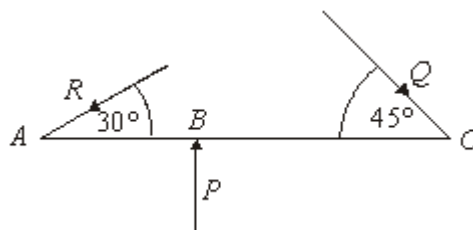
(Total 9 marks)

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Q40.

Three horizontal forces of magnitudes P , Q and R act on a light rod, ABC , which lies on a smooth horizontal table.

The force of magnitude P acts at B , at right angles to the rod, the force of magnitude Q acts at C , at an angle of 45° to the rod, and the force of magnitude R acts at A , at an angle of 30° to the rod, as shown in the diagram.



The three forces together are equivalent to a couple.

Given that $P = 4\sqrt{2}$, show that $R = 4(\sqrt{6} - \sqrt{2})$, and find Q , giving your answer in the form $a(b - \sqrt{c})$.

(Total 7 marks)

Q41.

The forces $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 3 \\ -3 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$ act in the x, y, z plane at the points with coordinates $(2, 5, 3)$, $(0, 11, 2)$ and $(5, -4, 7)$ respectively.

(a) (i) Find $\mathbf{F}$, the resultant of this system of forces.

(2)

(ii) Show that the magnitude of $\mathbf{F}$ is 13.

(2)

(b) The resultant of the two forces $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -3 \\ 5 \end{bmatrix}$ about the point $(3, 8, 7)$.

(i) Find the moment of the force $\begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$ about the point $(3, 8, 7)$.

(4)

(ii) Write down the moment of $\mathbf{F}$ about the point $(3, 8, 7)$. Explain your answer.

(2)

(Total 10 marks)

Q42.

Three forces act at points in the x, y plane as listed below:

$7\mathbf{i} + 2\mathbf{j}$	at	$(3, -5)$
$-\mathbf{i} + 4\mathbf{j}$	at	$(4, 1)$
$\mathbf{i} + 6\mathbf{j}$	at	$(8, 2)$

The three forces are equivalent to a single force, $\mathbf{F}$.

(a) Find the magnitude of $\mathbf{F}$.

(4)

(b) Show that the line of action of $\mathbf{F}$ acts through the point $\left(\frac{53}{6}, 0\right)$.

(5)

(Total 9 marks)

Q43.

The two forces $\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - 7\mathbf{j} - 6\mathbf{k}$, together with a third force, $\mathbf{F}$, form a couple.

Find:

- (a) the force $\mathbf{F}$;

(3)

- (b) the magnitude of $\mathbf{F}$.

(2)

(Total 5 marks)

Q44.

A force, $\mathbf{F}$, acts at R , the mid-point of a light rod PQ . The coordinates of P and Q are $(3, 4, 1)$ and $(-1, 6, 9)$ respectively and $\mathbf{F} = 7\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$. The three unit vectors, $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, are mutually perpendicular and in the directions of the x , y and z axes respectively.

Find the moment of the force $\mathbf{F}$ about the point P .

(Total 5 marks)

Q45.

Forces act at points in the x, y plane as listed below:

$5\mathbf{i}$	at	$(2, 2)$
$-6\mathbf{j}$	at	$(5, -1)$
$a\mathbf{i} - \mathbf{j}$	at	$(-7, 5)$
$2\mathbf{i} - \mathbf{j}$	at	$(b, 0)$

Given that this system is equivalent to a single force acting along the line $x = -1$, find the values of a and b .

(Total 7 marks)

Q46.

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Two forces, $3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$ and $2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$, act at the points whose coordinates are $(4, -2, 1)$ and $(5, 3, -4)$ respectively. The three unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are mutually perpendicular. The resultant of these forces, together with a force, $\mathbf{F}$, acting through the origin, form a couple.

Find:

- (a) (i) the force $\mathbf{F}$;

(2)

- (ii) the magnitude of $\mathbf{F}$;

(2)

- (b) the moment of the couple.

(5)

(Total 9 marks)

Q47.

Four forces act at the points with position vectors as listed in the table below, where a , b and c are constants.

Force		Position vector
$4\mathbf{i} - 3\mathbf{j}$	acting at	$\mathbf{0}$
$2\mathbf{i}$	acting at	$3\mathbf{j}$
$-\mathbf{j}$	acting at	$4\mathbf{i}$
$a\mathbf{i} + b\mathbf{j}$	acting at	$5\mathbf{i} + c\mathbf{j}$

The system is equivalent to a force $c(\mathbf{i} + \mathbf{j})$ acting at the origin together with an anti-clockwise couple of magnitude 34.

- (a) Show that the only two possible values that c can take are 3 and 8.

(8)

- (b) For each of these values for c , find the corresponding values of a and b .

(2)

(Total 10 marks)



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